



Exploratory Data Analysis Large-Scale Examination Dataset

A rigorous Python-based EDA of **2.33 million examination records** — uncovering performance patterns, regional disparities, and actionable benchmarks across 4,750 centers in India.

PORTFOLIO PROJECT

PYTHON · PANDAS · SEABORN

Made with GAMMA

Scale & Structure

2.33M

Records

Examination entries across India

4,750

Centers

Unique examination locations

51

States

Geographic coverage

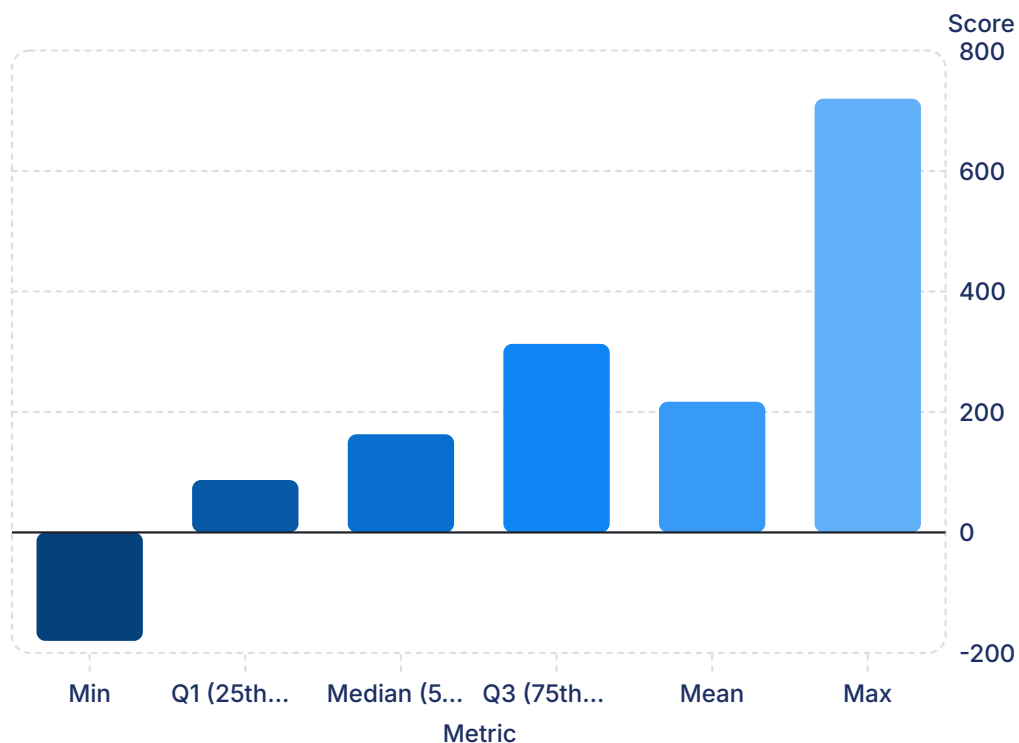
800

Score Range

-180 to 720

Six columns: **center_id**, **center_name**, **center_city**, **center_state**, **serial_no**, **score**. Memory footprint: **106.8 MB**. Zero duplicate records. Only **304 missing values** (0.013%) in **center_state**.

Score Distribution at a Glance



Key Observations

- **Mean (217) > Median (163)** — right-skewed distribution
- Std. deviation of **166** indicates wide performance spread
- 50% of students score between **87 and 313**
- Extreme range (-180 to 720) suggests negative marking or data anomalies

i The score range exceeds 900 points — warranting investigation into scoring rules or data quality issues.

UNIVARIATE ANALYSIS

Geographic Coverage

Top State by Volume

Uttar Pradesh — ~310K records, leading all states in examination participation

Top City by Volume

Delhi — ~63K records, the single highest city-level concentration

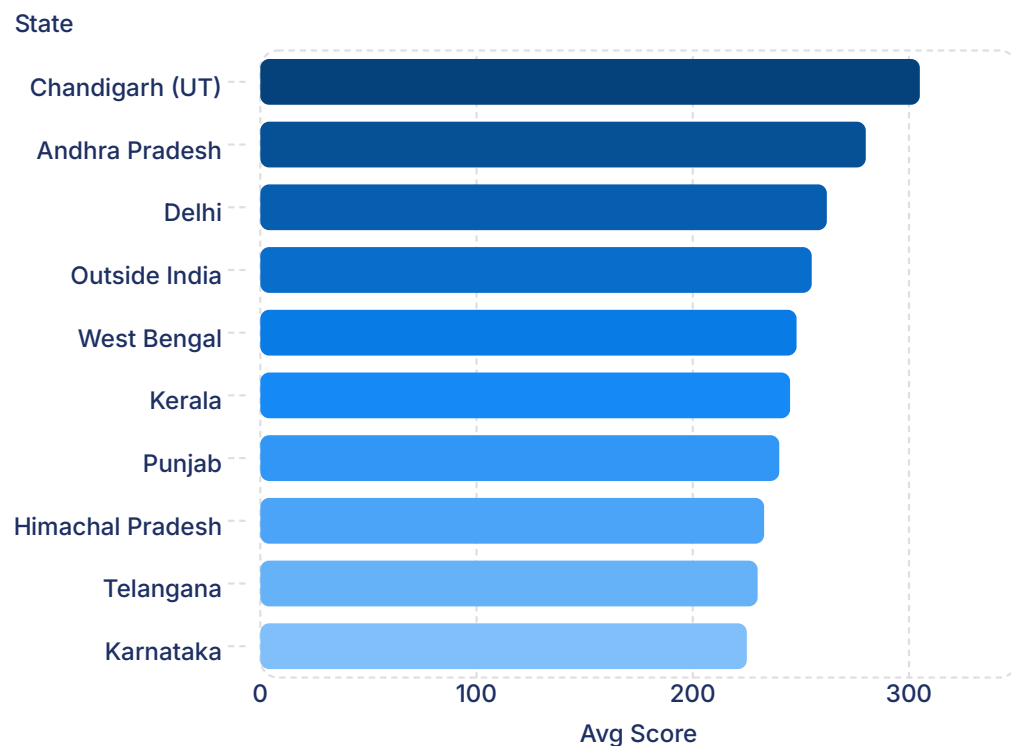
585 Cities

Spanning 51 states and union territories across India

Top 5 states by volume: **Uttar Pradesh (~310K)**, **Maharashtra (~275K)**, **Rajasthan (~190K)**, **Karnataka (~150K)**, **Tamil Nadu (~140K)**.



Average Score by State

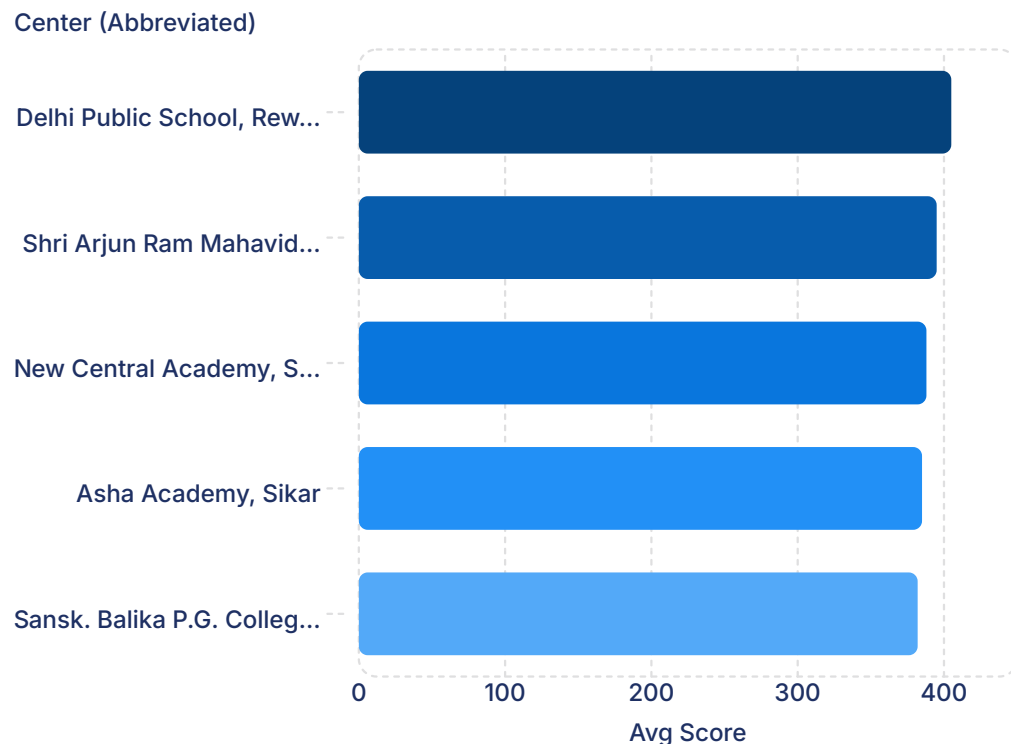


Regional Performance Gap

Chandigarh (UT) leads at **305** — 155 points above the lowest-ranked states (~150). Delhi, Andhra Pradesh, and Kerala form a strong second tier.

“Differences may be associated with regional educational and demographic factors, student preparation levels, and regional academic competitiveness. *Causality cannot be inferred from this data alone.*”

Top & Bottom Performing Centers



Key Findings

- **Top center:** Delhi Public School, Rewari — **405 avg**
- **Bottom centers:** Bottom centers show unusually low average scores, warranting further validation and investigation.
- **Sikar, Rajasthan** dominates the top 20 — a notable regional cluster
- Gap between top and bottom exceeds **400 points**



Bottom-performing centers show unusually low scores — investigate for data entry errors or incomplete records before drawing conclusions.

Data Quality & Outlier Detection

Quality Checks

Missing Values

Only **304 records** (0.013%) missing in center_state. All other columns complete.

Duplicates

Zero duplicate records detected across 2.33M rows.

Outliers (Z-Score)

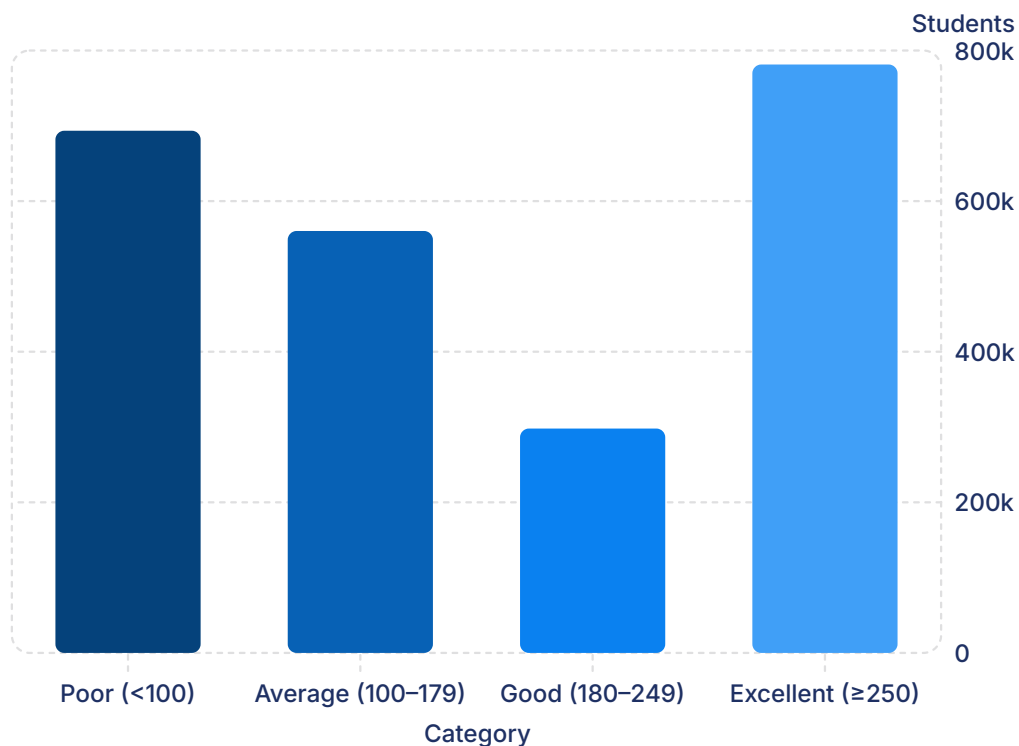
71 outliers identified using $|Z| > 3$ on score — ~0.003% of data.

Correlation Matrix

	center_id	serial_no	score
center_id	1.00	0.011	-0.004
serial_no	0.011	1.00	0.033
score	-0.004	0.033	1.00

All correlations are near zero — No meaningful linear relationship was observed between score and other numeric fields.

Score Categorization



Performance Segments

- **Excellent (≥ 250):** 781,415 students — **33.5%**
- **Poor (< 100):** 693,457 students — **29.7%**
- **Average (100–179):** 560,336 students — **24.0%**
- **Good (180–249):** 297,954 students — **12.8%**

Score segmentation reveals concentration at both lower and higher performance bands.



KEY INSIGHTS

What the Data Reveals

1 Regional Disparity is Significant

Chandigarh averages 305 vs. ~150 for lowest states — a 2× gap suggesting substantial regional performance variation..

2 Sikar, Rajasthan is an Outlier Cluster

Multiple centers from one district dominate the top 20 — warrants investigation into local academic ecosystems.

3 Polarized Score Distribution

33.5% Excellent and 29.7% Poor — highlighting uneven performance distribution across students.”

4 Data Quality is Strong

99.987% complete, zero duplicates, only 71 statistical outliers — a largely clean dataset with minimal missing values and no exact duplicates.”

EDA Project — Completed

Key Achievements

- Data cleaning & preprocessing
- Missing value & duplicate analysis
- Statistical & distribution analysis
- Outlier detection (Z-score)
- State & center benchmarking
- Feature engineering & segmentation

Tools Used

Python · Pandas · NumPy · Matplotlib · Seaborn · SciPy

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📄 This project demonstrates end-to-end EDA on a 2.33M-row dataset — from raw data to practical analytical insights